

Self Attention

The following formula describes the Attention Calculation in Transformers:

$$Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V$$

The sequence below is a vector representation of an input sequence of length = 3 and embedding dimension = 2:

$$seq = \begin{bmatrix} 1.5 & 0.5 \\ 1.5 & 0.0 \\ 0.5 & 0.5 \end{bmatrix}$$

The matrices below are the weights for query, key and value:

$$w_q = \begin{bmatrix} 2 & 2 \\ 2 & 0 \end{bmatrix}$$

$$w_k = \begin{bmatrix} 2 & 0 \\ 4 & 2 \end{bmatrix}$$

$$w_v = \begin{bmatrix} 2 & 4 \\ 0 & 2 \end{bmatrix}$$

Exercise 1:

Calculate the query Q and key K for all elements in the input sequence! Use these formulas:

$$Q = seq \times w_q \text{ and } K = seq \times w_k$$

Exercise 2:

Calculate the attention scores without scaling factor! The attention scores without scaling factor are calculated by this part of the Attention formula:

$$scores = Q \times K^T$$

Solution 1:

Q

$$\begin{bmatrix} 1.5 & 0.5 \\ 1.5 & 0.0 \\ 0.5 & 0.5 \end{bmatrix} \times \begin{bmatrix} 2 & 2 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 1.5 * 2 + 0.5 * 2 & 1.5 * 2 + 0.5 * 0 \\ 1.5 * 2 + 0.0 * 2 & 1.5 * 2 + 0.0 * 0 \\ 0.5 * 2 + 0.5 * 2 & 0.5 * 2 + 0.5 * 0 \end{bmatrix}$$
$$= \begin{bmatrix} 4 & 3 \\ 3 & 3 \\ 2 & 1 \end{bmatrix}$$

K

$$\begin{bmatrix} 1.5 & 0.5 \\ 1.5 & 0.0 \\ 0.5 & 0.5 \end{bmatrix} \times \begin{bmatrix} 2 & 0 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 1.5 * 2 + 0.5 * 4 & 1.5 * 0 + 0.5 * 2 \\ 1.5 * 2 + 0.0 * 4 & 1.5 * 0 + 0.0 * 2 \\ 0.5 * 2 + 0.5 * 4 & 0.5 * 0 + 0.5 * 2 \end{bmatrix}$$
$$= \begin{bmatrix} 5 & 1 \\ 3 & 0 \\ 3 & 1 \end{bmatrix}$$

Solution 2:

$$\begin{bmatrix} 4 & 3 \\ 3 & 3 \\ 2 & 1 \end{bmatrix} \times \begin{bmatrix} 5 & 3 & 3 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 * 5 + 3 * 1 & 4 * 3 + 3 * 0 & 4 * 3 + 3 * 1 \\ 3 * 5 + 3 * 1 & 3 * 3 + 3 * 0 & 3 * 3 + 3 * 1 \\ 2 * 5 + 1 * 1 & 2 * 3 + 1 * 0 & 2 * 3 + 1 * 1 \end{bmatrix}$$
$$= \begin{bmatrix} 23 & 12 & 15 \\ 18 & 9 & 12 \\ 11 & 6 & 7 \end{bmatrix}$$